Bangladesh Army University of Engineering & Technology

Department of CSE

EEE 1161

Summer 2024 Semester

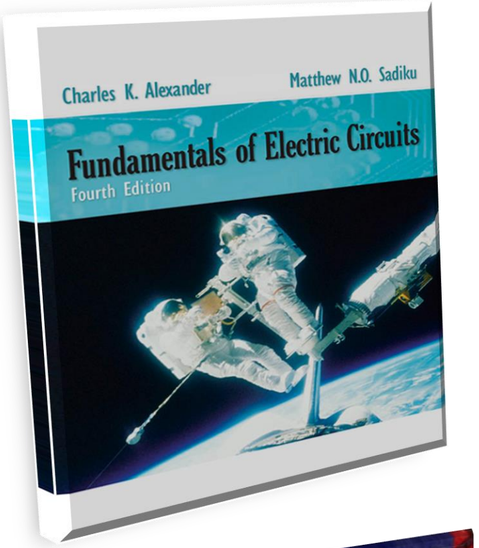
Lesson Plan

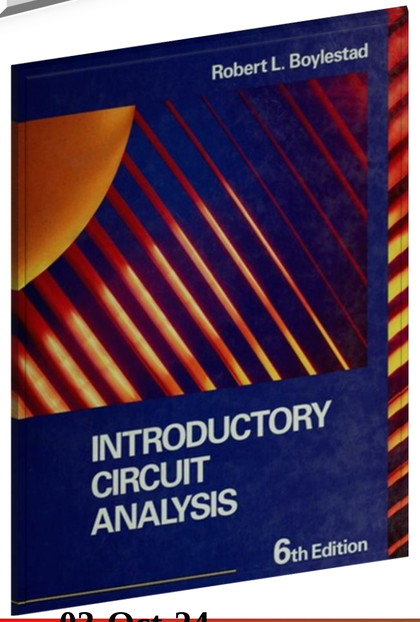
Lecture-1

Total Credit = 3.0
Credit Taking = 3.0

Course Teacher
Ashik Alam,
Asst., EEE, BAUET ,

Book References





Lecture Slide &

Reference Book:

1	**Fundamental of Electric Circuits**	**Charles K. Alexander and Mathew N. O. Sadiku (Fourth Edition)**
2	**Introductory Circuit Analysis**	**Robert L. Boylestad**
3	A Textbook of Electrical Technology volume 4	B. L Theraja

Introduction to sinusoidal signal(basic)

What is sinusoidal function?

- A sinusoid is a signal that has the form of the sine or cosine function.

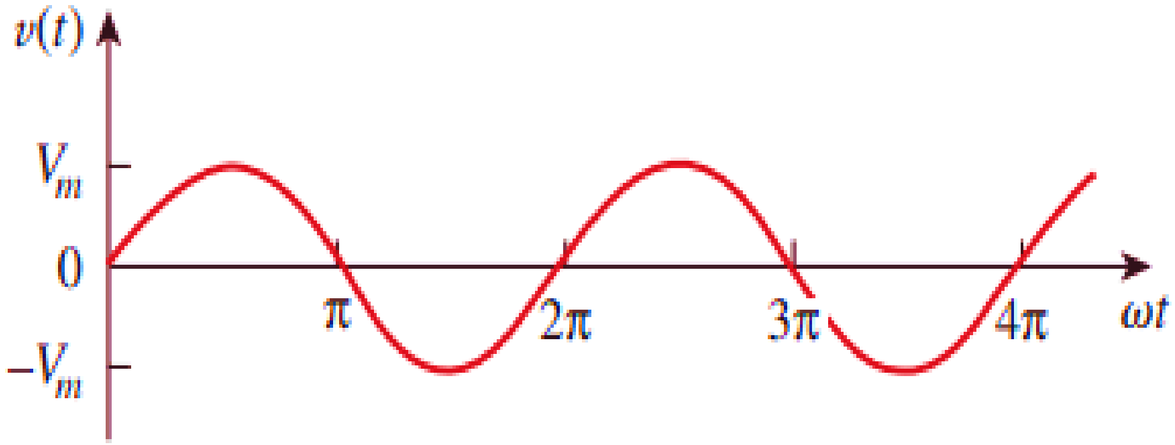
$$v(t) = VmSin\omega t$$

$$v(t+T) = VmSin\omega(t + \frac{2\pi}{T})$$

$$v(t+T) = VmSin\omega(t + 2\pi)$$

Hence,

$$v(t+T) = v(t)$$



$$T = \frac{2\pi}{\omega}$$

Introduction to sinusoidal signal(basic)

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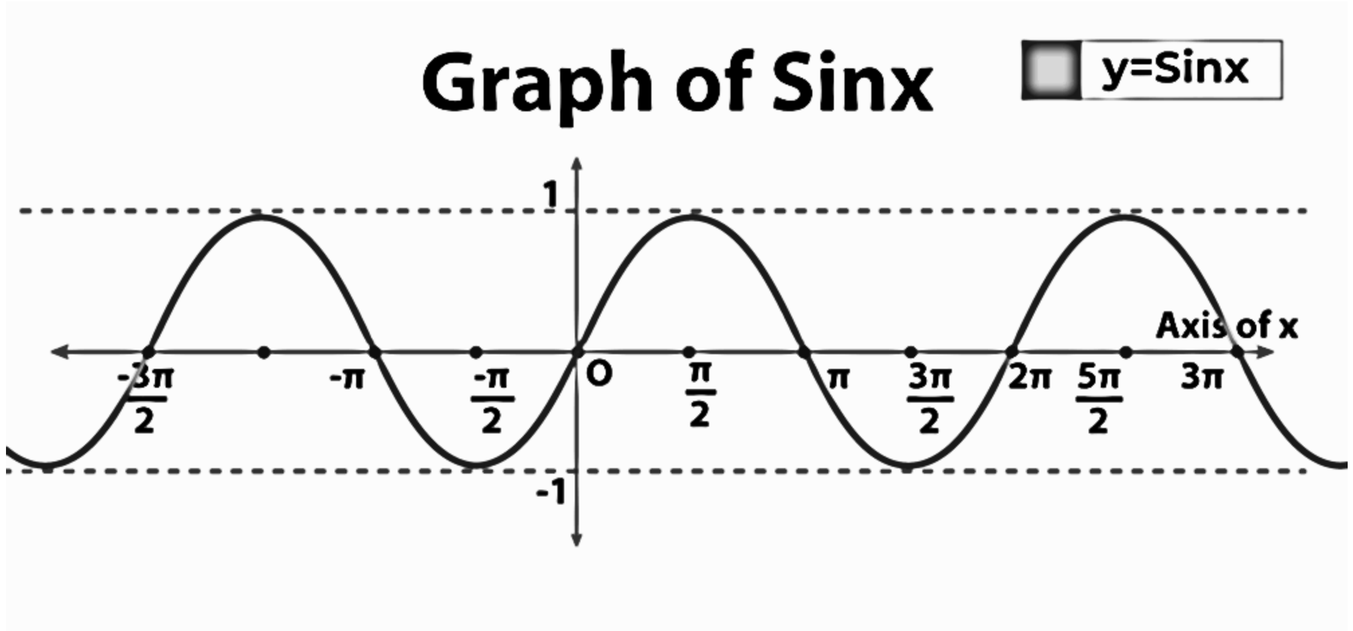

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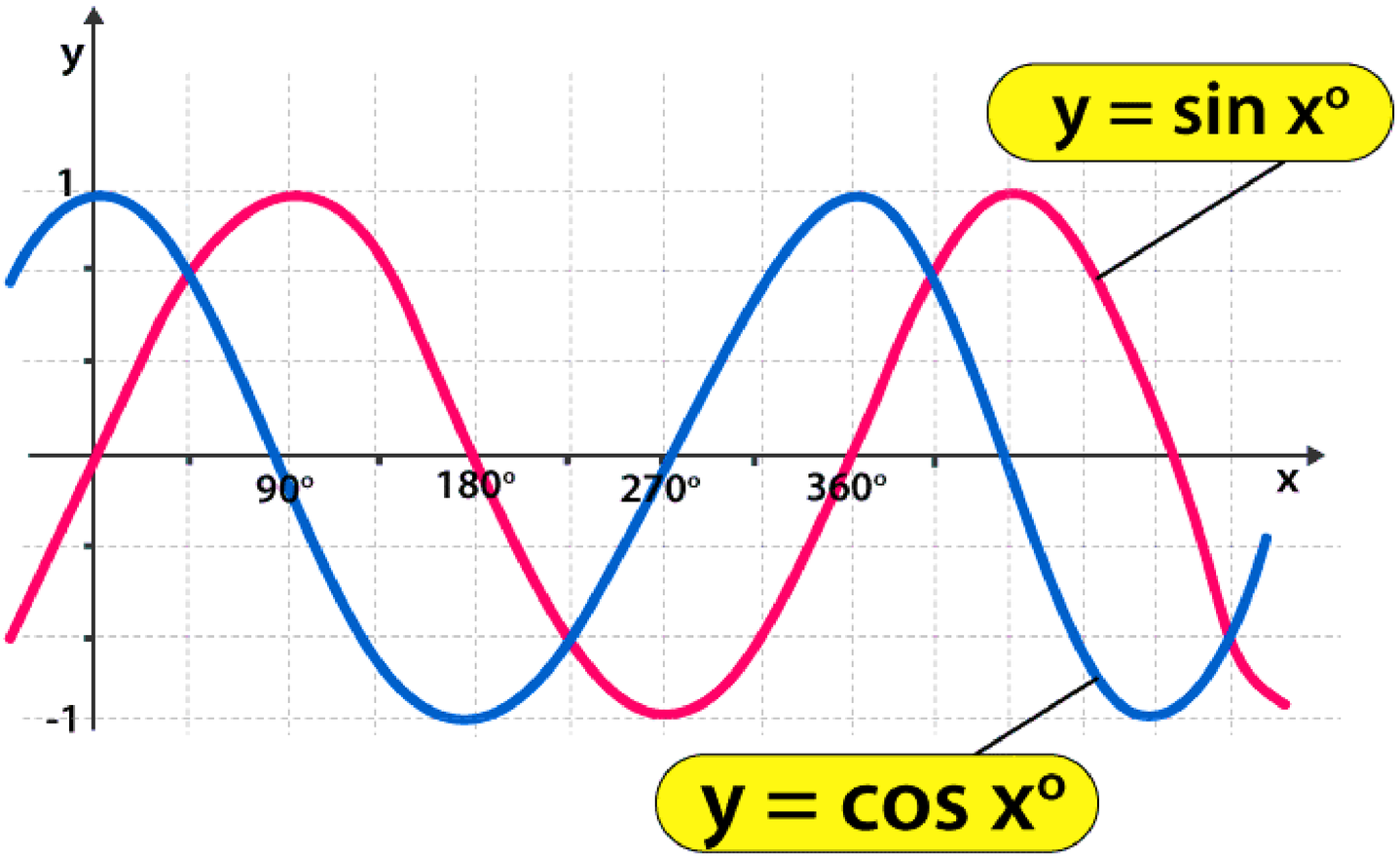

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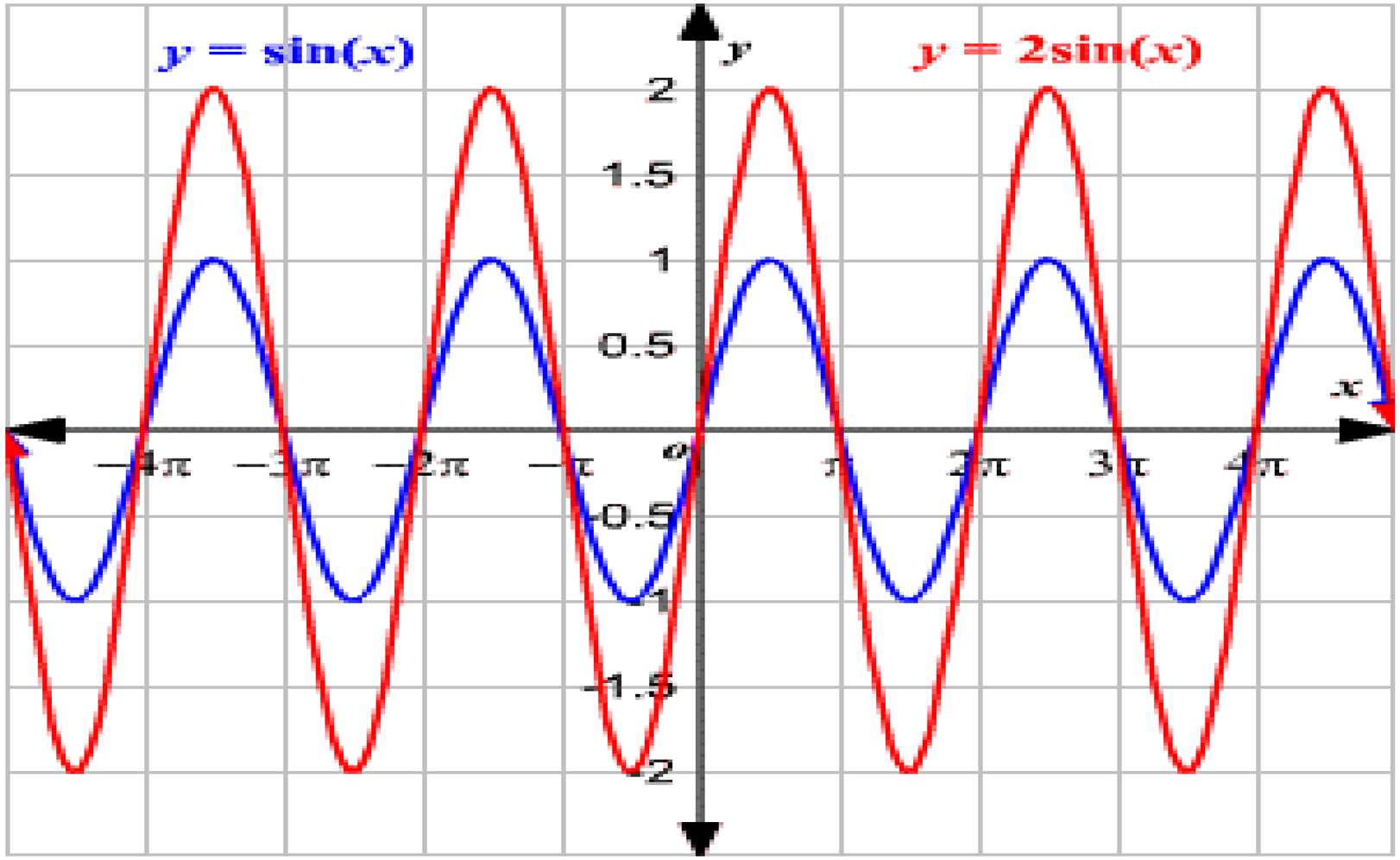

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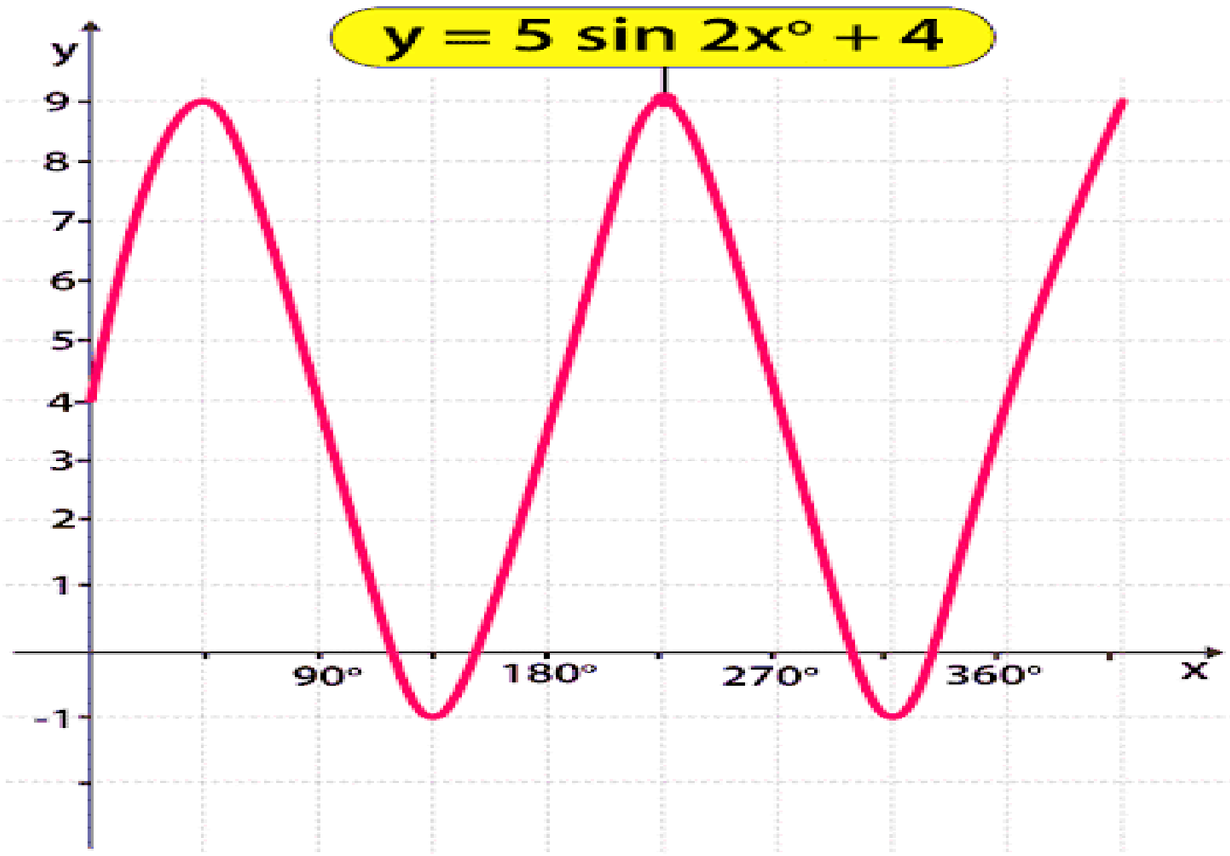

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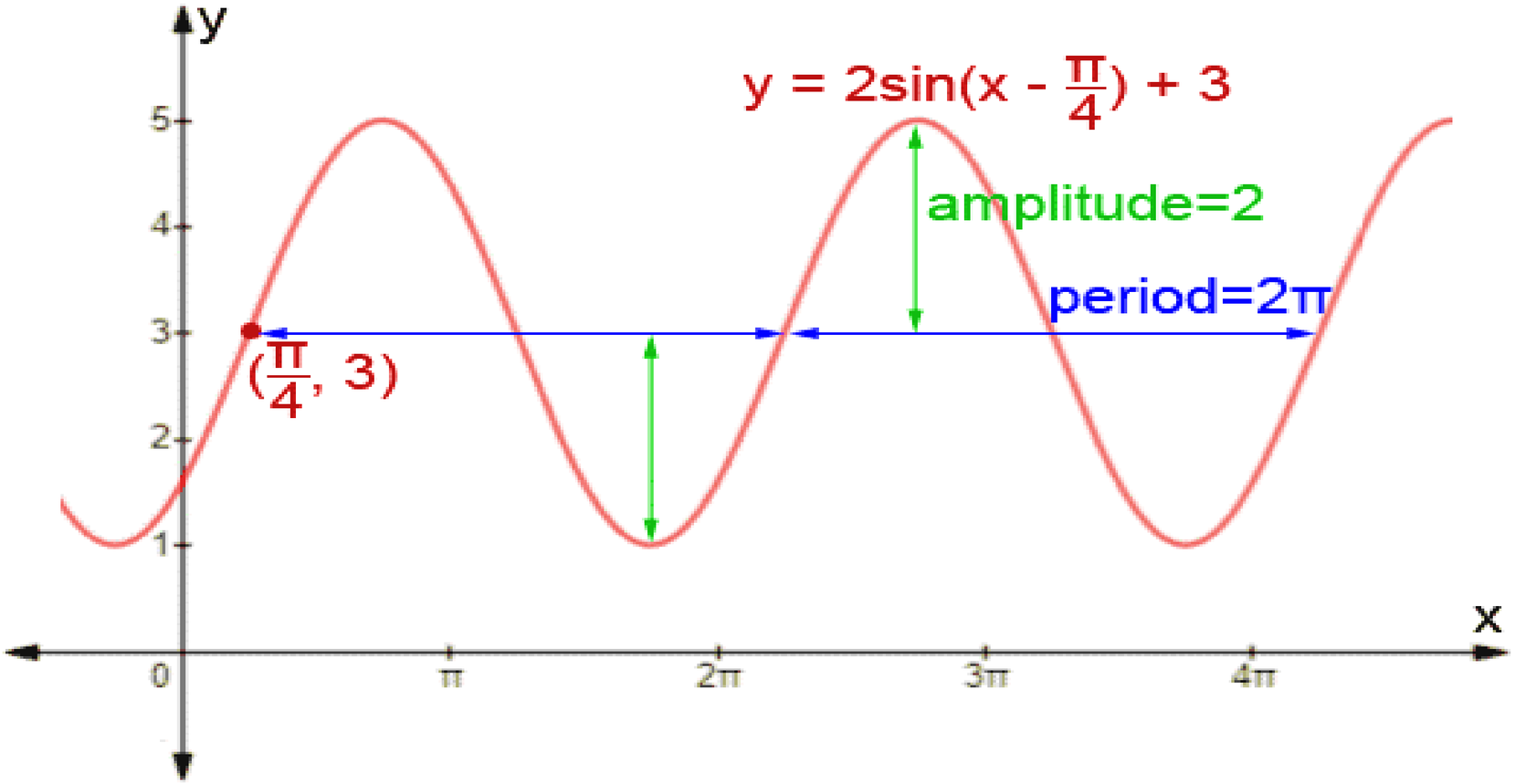


Example 1

- Find the amplitude, phase, period, and frequency of the sinusoid :

$$v(t) = 12\cos(50t + 10^{\circ})$$

Solution:

- The amplitude is $Vm = 12\ V$
- The phase is $\phi = 10^{\circ}$
- The angular frequency is $\omega = 50\ rad/s$.
- The period is $T = 2\pi/\omega = 2\pi/50 = 0.1257\ s$
- The frequency is $f = 1/T = 7.958\ Hz$

Example 2

- Calculate the phase angle between v1(t) and v2(t) and determine which sinusoid is leading.

Given two sinusoidal voltages:

$$v1(t) = 10\cos(100t + 30^{\circ})$$
$$v2(t) = 15\cos(100t + 75^{\circ})$$

Solution:

To compare v1(t) and v2(t), we must express them in the same form.

$$v1(t) = 10\cos(100t + 30^{\circ})$$
$$v2(t) = 15\cos(100t + 75^{\circ})$$

Example 2

$$\Delta\phi = 75^{\circ} - 30^{\circ} = 45^{\circ}$$

We can write v2(t) as:

$$v2(t) = 15\cos(100t + 75^{\circ})$$

$$= 15\cos(100t + 30^{\circ} + 45^{\circ})$$

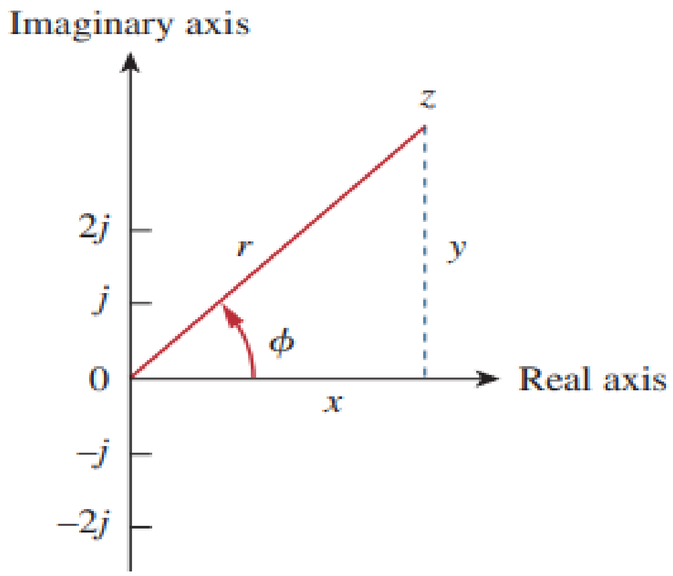
Comparing the expressions shows clearly that v2(t) leads v1(t) by 45°

Phasors

- phasor is a complex number that represents the amplitude and phase of a sinusoid.
- Phasors convert time-domain sinusoidal signals to the frequency domain.
- A sinusoidal voltage $v(t) = \mathrm{Vm}\cos(\omega t + \phi)$ can be represented as a phasor $V = Vme^{j\phi}$

- A complex number z can be written in rectangular form as z = x + jy
- The complex number z can also be written in polar or exponential form as $z = r\angle\varphi = re^{j\varphi}$

  where r is the magnitude of z, and φ is the phase of z



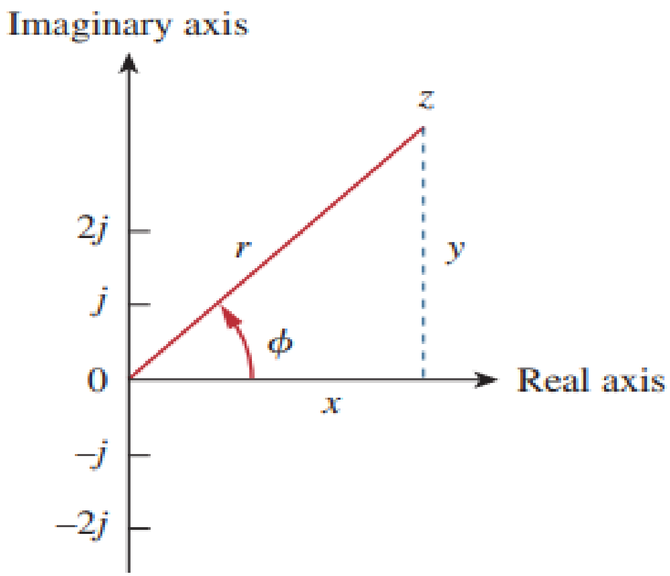
Phasors

- We notice that z can be represented in three ways:

$z = x + jy$ Rectangular form

$z = r \angle \varphi$ Polar form

$z = re^{j\varphi}$ Exponential form



z may be written as:

$$z = x + jy = r \angle \phi = r(\cos \phi + j \sin \phi)$$

complex Conjugate: $z* = x - jy = r\angle - \varphi = re^{-j\varphi}$ $[e^{\pm j\varphi} = \cos\varphi \pm j\sin\varphi]$

$$\cos\varphi = Re^{(j\varphi)}$$
$$\sin\varphi = Ime^{(j\varphi)}$$

Phasors

Some important operations of complex number: $z = x + jy = r\angle\phi$, $z_1 = x_1 + jy_1 = r_1\angle\phi_1$

$z_2 = x_2 + jy_2 = r_2\angle\phi_2$

Addition:

$$z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2)$$

Subtraction:

$$z_1 - z_2 = (x_1 - x_2) + j(y_1 - y_2)$$

Multiplication:

$$z_1 z_2 = r_1 r_2 \angle\phi_1 + \phi_2$$

Division:

$$\frac{z_1}{z_2} = \frac{r_1}{r_2}\angle\phi_1 - \phi_2$$

Reciprocal:

$$\frac{1}{z} = \frac{1}{r}\angle -\phi$$

Square Root:

$$\sqrt{z} = \sqrt{r}\angle\phi/2$$

Complex Conjugate:

$$z^* = x - jy = r\angle -\phi = re^{-j\phi}$$

The idea of phasor representation is based on Euler's identity.

$$e^{\pm j\emptyset} = cos\emptyset \pm j\, sin\emptyset$$

$$\cos\phi = \mathrm{Re}(e^{j\phi})$$

$$\sin\phi = \mathrm{Im}(e^{j\phi})$$

Phasors

$$v(t) = V_m \cos(\omega t + \phi) = \mathrm{Re}(V_m e^{j(\omega t + \phi)})$$

$$v(t) = \mathrm{Re}(V_m e^{j\phi} e^{j\omega t})$$

$$\boxed{v(t) = \mathrm{Re}(\mathbf{V} e^{j\omega t})}$$

$$\mathbf{V} = V_m e^{j\phi} = V_m \angle \phi$$

$$\frac{dv}{dt} = -\omega V_m \sin(\omega t + \phi) = \omega V_m \cos(\omega t + \phi + 90^\circ)$$

$$= \mathrm{Re}(\omega V_m e^{j\omega t} e^{j\phi} e^{j90^\circ}) = \mathrm{Re}(j\omega \mathbf{V} e^{j\omega t})$$

Sinusoid-phasor transformation.

Time domain representation	Phasor domain representation
$V_m \cos(\omega t + \phi)$	$V_m \angle \phi$
$V_m \sin(\omega t + \phi)$	$V_m \angle \phi - 90^\circ$
$I_m \cos(\omega t + \theta)$	$I_m \angle \theta$
$I_m \sin(\omega t + \theta)$	$I_m \angle \theta - 90^\circ$

$$\underset{\text{(Time domain)}}{\frac{dv}{dt}} \Leftrightarrow \underset{\text{(Phasor domain)}}{j\omega \mathbf{V}}$$

$$\underset{\text{(Time domain)}}{\int v\, dt} \Leftrightarrow \underset{\text{(Phasor domain)}}{\frac{\mathbf{V}}{j\omega}}$$

Phasors

The differences between *v* and **V** should be emphasized:

1. *v* is the *instantaneous or time domain* representation, while **V** is the *frequency or phasor domain* representation.

2. *v* is time dependent, while **V** is not. (This fact is often forgotten by students.)

3. *v* is always real with no complex term, while **V** is generally complex

Try to Do this!

- Question 1

Transform these sinusoids to phasors:

(a) $i = 6\cos(50t - 40^\circ)$ A
(b) $v = -4\sin(30t + 50^\circ)$ V

- Question 2

Find the sinusoids represented by these phasors:

(a) $\mathbf{I} = -3 + j4$ A
(b) $\mathbf{V} = j8e^{-j20^\circ}$ V

- Question 3

Using the phasor approach, determine the current $i(t)$ in a circuit described by the integrodifferential equation

$$4i + 8\int i\,dt - 3\frac{di}{dt} = 50\cos(2t + 75^\circ)$$

Related Questions

- ✓ What is a phasor? What is the relationship between angular frequency and frequency?
- ✓ Explain the significance of the phase angle in a sinusoidal function.
- ✓ Describe the process of multiplying two phasors.
- ✓ Given $v1(t) = 10\cos(100t + 30^{\circ})$ and $v2(t) = 15\cos(100t + 75^{\circ})$, calculate the phase angle difference and determine which sinusoid is leading.
- ✓ $v(t) = 20\sin(50t + 60^{\circ})$ to its phasor form.

Thank you